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A Digital Logic Design course introduces fundamental concepts essential for understanding and designing digital systems. Students learn about logic gates, Boolean algebra, and design methodologies for both combinational and sequential circuits. Practical applications include designing arithmetic circuits, memory units, and more, preparing students for roles in computer engineering and related fields. Through hands-on labs and projects, they develop critical thinking, problem-solving, and teamwork skills, essential for tackling real-world challenges in digital system design and optimization.

This course is concerned with the logic design of digital electronic circuits. Digital circuits are employed in the design of digital systems such as digital computers, control systems, data communications, and many other applications that required electronic digital hardware. The course covers the basic tools for designing digital circuits and teaches methods and procedures suitable for a variety of digital applications.

Upon successful completion of this course, students will be able to:

18. Mapping of CO-PO-WK-WP-WA:

#	COs	POs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CO1	Underst and the knowled ge of Number system, Boolean algebra and different types of Combin ational and Sequenti al logic circuits.	PO1	C2	K1, K5	-	-	Lecture, Class discussion, Note	Exam, Quiz, Assignment
CO2	Interpret different digital electronics circuits in terms of different systems of Boolean expression and their simplification, truth table, state table etc.	PO2	C3	K3	-	-	Discussion, Assignment	Assignment, Quiz, Exam
CO3	Analyze various problems related to digital electronics and implement digital circuits like adder, comparator, converter, decoder, encoder, ROM, PLA, counter, register etc.	PO3	C4	K6	-	-	Lecture, Assignment, Quiz	CT, Assignment

Note: **DL:** Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities
Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)

19. Course Content

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Binary Systems, Boolean Algebra and Logic Gates: Number system, binary codes, binary logics, logic gates, Boolean algebra, canonical and standard forms.	2 weeks	CLO1
2	Simplification of Boolean Functions: The Map Method. Two-, Three-, Four-, Five And Six- variable Maps, Product of Sum Simplification, NAND And NOR Implementation, Don't Care Conditions, Multilevel NAND Circuits, Multilevel NOR Circuits, Exclusive-or and Equivalence Functions.	2 weeks	CLO1, CLO2
3	Combinational Logic: Design Procedure, Adders, Subtractors, Code Conversion, Analysis Procedure, designing various types of combinational circuit using logic gates.	2 weeks	CLO2
Final Exam: 50 Marks Group-A (20 Marks)			
4	Combinational logic with MSI and LSI: MSI and LSI, Binary Parallel Adder, look ahead carry, decimal Adder, Magnitude Comparator, decoder, encoder, multiplexer & demultiplexer	1 week	CLO2
5	Sequential Logic: Flip-flops, triggering of flip-flops, analysis of clocked Sequential circuits, state reduction and Assignment, design procedure, design with state equations, designing various types of sequential circuits	3 weeks	CLO2
Group-B (30 Marks)			
6	Digital Integrated Circuits: Bipolar transistor characteristics, RTL, DTL, Integrated-Injection logic, emitter couple logic, metal-oxide semiconductor, complementary MOS.	1 week	CLO2
7	Registers, Counters, and the Memory: Registers, shift registers, ripple counters, asynchronous counter, and synchronous counter, memory, read only memory, programmable logic array, random access memory, and memory unit.	2 weeks	CLO3
8	Register Transfer and Processor Logic Design: Interregister transfer, arithmetic-logic and shift-operations, design a simple computer, processor organization, arithmetic logic unit, design of	2 week	CLO3

	arithmetic logic unit, design of accumulator.		
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20. Weekly Activity Plan:

Week	Topic	TLS	AS	CLOs
Week 1	Introduction: Digital Systems, Binary Numbers, Number base conversions, Complements, Binary codes and binary logic, Boolean Algebra, Boolean functions	Lecture	CT, Mid	CLO1
Week 2	Canonical and standard forms, Other logic operations, Digital logic gates.	Lecture, Discussion	Class work, Assignment	CLO1
Week 3	The k-map method, Two-variable map and Three-variable map, Four and Five variable map	Lecture, Discussion	Class work, Quiz	CLO2
Week 4	Product of sums simplification, Don't-care conditions	Lecture, Discussion	Class work, CT, Quiz, Assignment	CLO2
Week 5	NAND and NOR implementation Other Two-level implementations Exclusive – OR function.	Lecture, Discussion	CT, Assignment	CLO2
Week 6	Review of the midterm exam content			
Week 7	Combinational circuit analysis procedure and design procedure, Binary Adder-Subtractor Multilevel NAND circuits and Multilevel NOR circuits.	Lecture, Lab work, Assignment	Assignment, quiz	CLO2
Week 8	Binary Parallel Adder, BCD Adder Magnitude Comparator.	Lecture, Lab work, Assignment	Assignment, CT	CLO2
Week 9	Decoders, Encoders, Multiplexers, De-Multiplexers	Lecture, Lab work, Assignment	Assignment, Quiz	CLO2
Week 10	Latches, Flip-Flops	Lab	CT, Lab work	CLO2
Week 11	Analysis of clocked sequential circuits, Design of synchronous sequential circuits	Lecture, Lab work, Assignment	Assignment, CT	CLO2
Week 12	Registers, shift registers, Ripple counters	Lecture, Discussion	Class work, Assignment	CLO3
Week 13	Synchronous counters-1, Synchronous counters-2	Lecture, Lab work, Assignment	Assignment, CT	CLO3

Week	Topic	TLS	AS	CLOs
Week 14	Random-Access Memory,Read-only memory-1 Read-only memory-2, Programmable logic Array	Lecture, Lab work, Assignment	Assignment, CT	CLO3
Week 15	Review of the semester final content			

Part C

21. Assessment Strategy:

The assessment and evaluation strategies for the course are given as follows:

Assessment Strategies	Description
Class Participation:	Students' individual in-class responses, attention, and sense of discipline, morality will be judged on the basis of 10 (ten) marks.
Class Test/Quiz:	Students will sit for 2 class test and 1 quiz during the semester. The test/quiz will be taken before midterm. Class test/quiz marks will be assessed in 10(five). No makeup class test will be taken. Students are strongly recommended not to miss any test.
Assignment:	No late submission of assignments will be accepted.
Midterm Exam:	Midterm exam will be held according to the Academic Calendar published by the university. Midterm assessment marks will be 30 (thirty).
Final Exam:	Final exams will be held according to the Academic Calendar published by the university. Final assessment marks will be 40 (forty). Course contents learnt before Midterm examination will be included in the syllabus of Final Examination.
Make-up Procedure:	No late submission and/or make-up assignment/presentation/quiz will be allowed without prior permission and adequate and reasonable proof of absence.

22. Marks Distribution:

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Mid-term (30)	Written Exam (50)
Remember				5	5
Understand			5	10	15
Apply			5	15	30
Analyze					
Evaluation					
Create					
	Responding	10			
Total allocated marks		10	10	30	50

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, Discussions Powerpoint Presentation,

Assessment tools: Class Attendance, Class test, Quizzes/ Assignment on problem solution, Mid-Term & Final Exam. Project evaluation & Viva

23. Grading Policy: As per IIUC grading policy

24. Code of Conduct:

- a. It is mandatory for all the students to participate in the class regularly and maintain proper discipline in the class.
- b. If a student fails to attend any class test, term exams, or final examination, he/she will get a zero in that class test, term, or final examination.
- c. Adopting unfair means in the exams will be considered as a serious crime and the student shall be placed to the university disciplinary committee.
- d. All the assignments, class test and exam copies should be neat and clear and demonstrate professionalism.
- e. No student is allowed to duplicate other student's work directly or with minor changes.
- f. Plagiarism is strictly restricted. One needs to provide a reference while using someone else's words, ideas, or research in assignments/exams.

Part D

25. Learning Materials:

Text Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	M. Morris Mano	Digital Logic and Computer Design	5 th Edition	Pearson education	2013	978- 0-13-277421-5

Reference Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	Md.Mozammel Huq Azad Khan	Digital logic design	1st edition	University Grants Commission of Bangladesh	2006	978-9848090244
2	Thomas L. Floyd	Digital Fundamentals	11 th Edition	Pearson	2014	978-0132737968
3	Tocci-widmer	Digital Systems	12th Edition	Pearson Prentice Hall	2016	978-0134220130